is a format developed by Esri and is most commonly found in desktop GIS applications. But the most human-readable form of geodata is latitude and longitude (or xy coordinates like we list in our file) when dealing with points like cities, oftentimes in a CSV. We'll use cities.csv, shown in the following listing. This is the same CSV we measured in chapter 2 that had the locations of eight cities from around the world.

Listing 7.2 cities.csv

```
"label","population","country","x","y"
"San Francisco", 750000,"USA",-122,37
"Fresno", 500000,"USA",-119,36
"Lahore",12500000,"Pakistan",74,31
"Karachi",13000000,"Pakistan",67,24
"Rome",2500000,"Italy",12,41
"Naples",1000000,"Italy",14,40
"Rio",12300000,"Brazil",-43,-22
"Sao Paolo",12300000,"Brazil",-46,-23
```

One thing you'll notice is that the latitudes and longitudes are imprecise. San Francisco, for instance, isn't at 37,-122 but rather 37.783, -122.417. When you plot these cities, they're going to look pretty off as you zoom in. Obviously, you'll want to use more accurate coordinates for your maps, but for this example, which mostly uses maps that are zoomed way out, this should be fine.

If you only have city names or addresses and need to get latitude and longitude, you can take advantage of geocoding services that provide latitude and longitude from addresses. These exist as APIs and are available on the web for small batches. You can see an example of these services maintained by Texas A&M at http://geoservices.tamu.edu/Services/Geocode/.

When dealing with more complex geodata like shapes or lines, you'll necessarily deal with more complex data formats. You'll want to use GeoJSON, which has become the standard for web-mapping data.

GeoJSON

GeoJSON (geojson.org) is, like it sounds, a way of encoding geodata in JSON format. Each *feature* in a *featureCollection* is a JSON object that stores the border of the feature in a *coordinates* array as well as metadata about the feature in a *properties* hash object. For instance, if you wanted to draw a square that went around the island of Manhattan, then it would have corners at [-74.0479, 40.6829], [-74.0479, 40.8820], [-73.9067, 40.8820], and [-73.9067, 40.6829], as shown in figure 7.2. You can easily export shapefiles into GeoJSON using QGIS (a desktop GIS application; qgis.org), PostGIS (a spatial database run on Postgres; postgis.net), GDAL (a library for manipulation of geospatial data; gdal.org), and other tools and libraries.

A rectangle drawn over a geographic feature like this is known as a *bounding box*. It's often represented with only two coordinate pairs: the upper-left and bottom-right corners. But any polygon data, such as the irregular border of a state or coastline, can be represented by an array of coordinates like this. In the following listing, we have a